Entertainment Planning Assistant Agent

Course Name: Agentic AI

Institution Name: Medicaps University – Datagami Skill Based Course

Student Name(s) & Enrolment Number(s):

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Project Number: *AAI-34*

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Academic Year: 4th year

Problem Statement & Objectives

1. Problem Statement
2. Project Objectives
3. Scope of the Project

Proposed Solution

1. Key features *(Just mention key features here no need to go into details)*
2. Overall Architecture / Workflow
3. Tools & Technologies Used (*If applicable)*

Results & Output

Add the below details here:

1. *Screenshots / outputs*
2. *Reports / dashboards / models*
3. *Key outcomes*

Conclusion

Mention a brief conclusion about your project summing up everything you have worked on and the key learning.

Future Scope & Enhancements

📌 Problem Statement & Objectives

Problem Statement

Planning entertainment-related activities such as movie analysis, reviews, or event organization involves multiple steps like research, validation, scheduling, and execution. Manual handling of these tasks is often inefficient, unstructured, and time-consuming, especially when dependencies between tasks exist.

There is a need for an intelligent system that can **automatically plan, organize, and optimize entertainment workflows** with minimal human intervention.

Project Objectives

- Design an intelligent **Planner Agent** capable of handling complex entertainment tasks.
- Enable **automatic decomposition** of high-level goals into actionable steps.
- Implement a **multi-step reasoning loop** for structured planning.
- Simulate **resource validation** using mock tools/interfaces.
- Generate an **optimized execution schedule** considering dependencies.
- Improve efficiency, scalability, and automation in entertainment planning tasks.

📌 Scope of the Project

- Focuses on **entertainment domain tasks** (e.g., movie reviews, event planning).
- Supports **goal-based planning** rather than manual step-by-step execution.
- Uses **mock tools/APIs** (not real-time production integrations).
- Limited to **planning and scheduling**, not full execution automation.
- Can be extended to other domains like education, business workflows, etc.

📌 Proposed Solution

The solution is an **AI-powered Planner Agent** that:

1. Accepts a **high-level user goal** (e.g., "Analyze popular movies").
2. Break it down into **smaller, logical subtasks**.
3. Validates each step using **mock resource-checking tools**.
4. Organizes tasks into a **dependency-aware execution sequence**.
5. Outputs a **structured plan and schedule**.

This agent mimics real-world decision-making by combining:

- Task decomposition
- Resource validation
- Dependency handling
- Workflow optimization

🚀 Key Features

- Goal-based task planning
- Multi-step reasoning loop
- Task decomposition engine
- Dependency-aware scheduling
- Mock tool/resource validation
- Dynamic plan generation
- Modular and scalable design

🏗️ Overall Architecture / Workflow

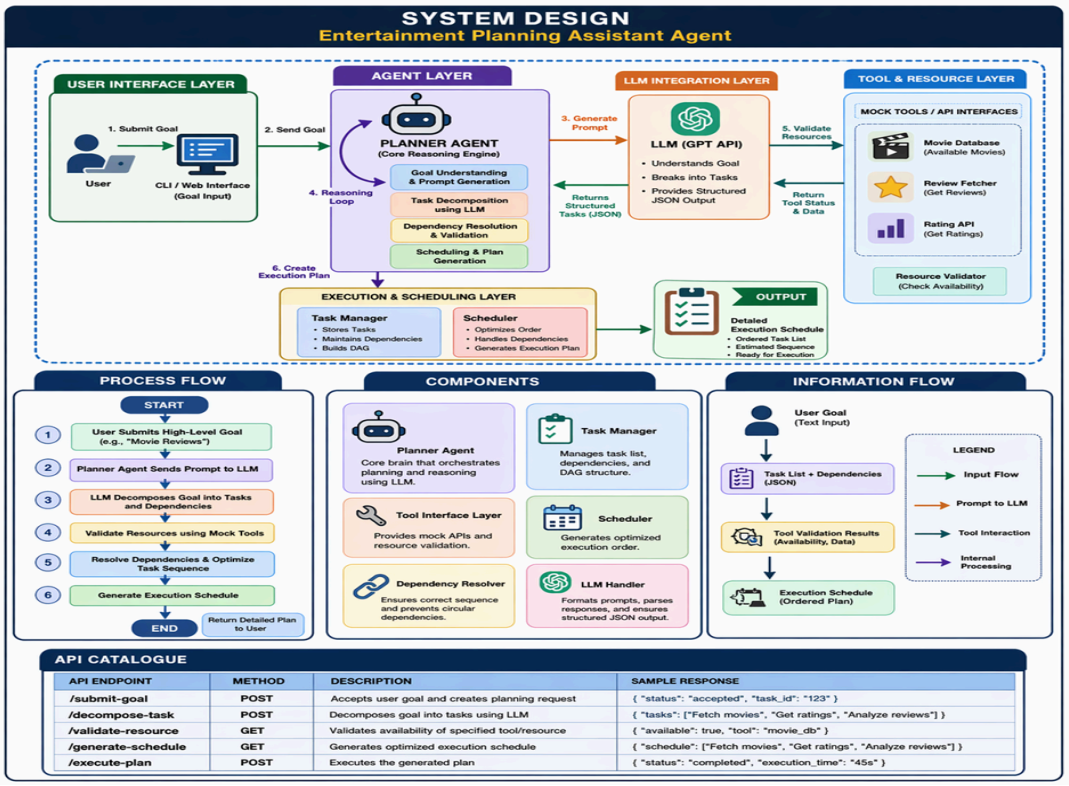
Workflow Steps:

1. **User Input**
    - High-level goal (e.g., "Movie Reviews")

2. **Planner Module**
    - Breaks goal into tasks
    - Identifies dependencies
3. **Reasoning Engine**
    - Iteratively refines tasks
    - Ensures logical sequencing
4. **Tool Validation Layer**
    - Simulates resource availability
    - Checks feasibility
5. **Scheduler**
    - Orders tasks efficiently
    - Optimizes execution flow
6. **Output Generator**
    - Produces final execution plan

Architecture Components:

- Input Interface (Streamlit UI)
- Planner Agent Core
- Reasoning Loop Engine
- Mock Tool APIs
- Task Scheduler
- Output Module



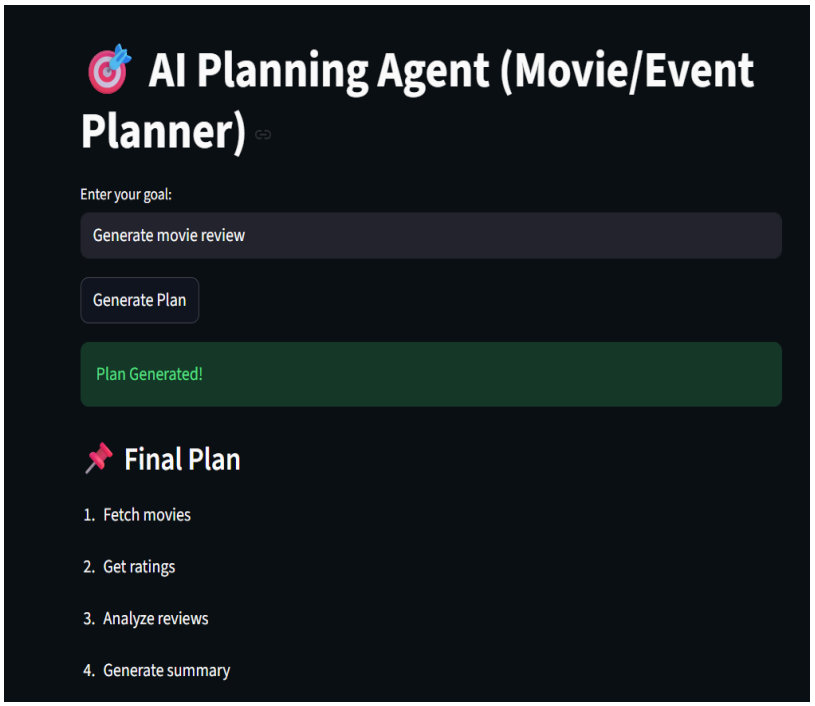
API ENDPOINT	METHOD	DESCRIPTION	SAMPLE RESPONSE
/submit-goal	POST	Accepts user goal and creates planning request	{ "status": "accepted", "task_id": "123" }
/decompose-task	POST	Decomposes goal into tasks using LLM	{ "tasks": ["Fetch movies", "Get ratings", "Analyze reviews"] }
/validate-resource	GET	Validates availability of specified tool/resource	{ "available": true, "tool": "movie_db" }
/generate-schedule	GET	Generates optimized execution schedule	{ "schedule": ["Fetch movies", "Get ratings", "Analyze reviews"] }
/execute-plan	POST	Executes the generated plan	{ "status": "completed", "execution_time": "45s" }

🛠️ Tools & Technologies Used

- **Python** – Core development
- **Streamlit** – User Interface
- **Ollama (TinyLLaMA / other models)** – LLM-based reasoning
- **JSON** – Task structure and communication
- **Requests API** – Model communication

📊 Results & Output

Screenshots / Outputs



- Streamlit UI showing:
  - User input goal
  - Generated task plan
- Console/JSON output:
  - Step-by-step execution plan

Reports / Dashboards / Models

- Structured task list (JSON format)
- Execution flow visualization (optional)
- Model-generated reasoning outputs

Key Outcomes

- Successfully generated **automated plans** from high-level goals
- Demonstrated **multi-step reasoning capability**
- Achieved **dependency-aware task sequencing**
- Built a **scalable agent-based architecture**
- Reduced manual effort in planning tasks

📌 Conclusion

The **Entertainment Planning Assistant Agent** demonstrates how intelligent systems can automate complex planning tasks using AI-driven reasoning. The project successfully integrates task decomposition, validation, and scheduling into a unified workflow.

It highlights the power of **agent-based systems** in handling real-world problems where multiple steps, dependencies, and decision-making processes are involved. This project also strengthened understanding of **LLMs, system design, and workflow automation**.

🔮 Future Scope & Enhancements

- Integration with **real APIs** (movie databases, event platforms)
- Upgrade to **more powerful LLMs** for better reasoning
- Add **execution capability** (not just planning)
- Implement **real-time collaboration features**
- Add **visual dashboards** for task tracking
- Improve **UI/UX design**
- Extend to **multi-domain planning** (education, business, travel)